

Innovation Insights Quarterly: Q3 2025

Date: 2026-01-28

Source: <https://www.franklintempleton.com/articles/2025/equity/innovation-insights-quarterly-q3-2025>

Extraction method: selenium

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We have noted before that the investment industry loves to categorize, whether by asset classes, capitalization buckets, industry sectors or style boxes. Yet we have found that innovation abhors a label—it consistently escapes attempts to define or corner it. Found throughout the economy, innovation is by its nature protean, connecting all disciplines and sectors. Our selections for this quarter capture this spirit—they collapse the boundaries between organic and artificial, illusion and reality, and human and machine.

Lasers create a brand-new color

Scientists used lasers to activate certain cells inside the human retina that then produced the perception of a new color.¹ Named “Olo,” the color is an intensely saturated blue-green, likened to peacock blue or teal. To achieve this effect, lasers controlled the amount of light delivered to each cell, tricking the part of the brain that processes vision to see an entirely new color.

Why it matters: On its own, creating a new color is an impressive technical achievement, especially because it can be repeated to make even more new colors. Beyond this, however, this technology could provide a way to treat color-blindness, a disease that affects as many as 350 million people worldwide.² Today, color-blind people are restricted from work as pilots, firefighters, doctors, and other professions.³

Artificial whale feces are deployed to re-enrich the ocean

Back when households demanded whale blubber to power their lamps, hunters killed 99% of the ocean’s whales. In the process, they accidentally eliminated one of the ocean’s most important nutrient recyclers: whale feces. Whales descend deep into the ocean, eat nitrogen-rich food, and then return to the surface and disperse that nutrient in their excrement. Without this natural cycle, the marine ecosystem has suffered. Thankfully, researchers have now devised a synthetic substitute to revitalize current “dead zones” in the ocean where no life can currently survive.⁴

Why it matters:

By reintroducing a whale feces proxy, researchers believe they may be able to restore ocean wildlife to its level from 400 years ago within the next 40 or 50 years. Further, healthy marine ecosystems naturally sequester carbon in the form of phytoplankton on the sea floor. This innovation has the potential to remove as much as 4% of total carbon emissions from the atmosphere per year.⁵

Exhibit 1: Organic Buildings of the Future?

The buildings of the future could be formed out of organic material like fungi, programmed to shape themselves into structures.

For illustrative purposes only.

Immunotherapy cures a golden retriever’s lung cancer

Lola, a golden retriever, was diagnosed with an aggressive lung cancer and given six months to live. Her owner enrolled her in a clinical trial at UC Davis in California to try a novel immunotherapy. Alongside radiation, she was administered the experimental cancer-fighting medication via inhalation. Today, Lola has been cancer-free for 2 years.⁶

Why it matters: Cancer is strikingly similar in dogs and humans. Dogs share 85% of our DNA and have similar microbiomes and immune systems, but they are 10x more likely to get cancer.⁷ When they have the disease, their tumors grow much faster, which makes the cancer easier to study. Further, there are fewer restrictions on tumor gene sequencing, data analysis, and experimental treatment plans for dogs. All of this combines to make “comparative oncology,” which uses dogs as models for humans, a highly promising area of cancer research.

Brain cells and silicon are fused to make world’s first biological computer

An Australian company recently connected human neurons with silicon hardware to make a computer that far exceeds the performance of existing silicon-based artificial intelligence (AI) chips. To make this biological computer, lab-grown neurons are placed on an array of electrodes inserted into a rectangular life-support unit and connected to a software-based system that operates in real-time.⁸

Why it matters:

This cell network on a silicon chip is a dynamic and sustainable organic computer that can learn so quickly and flexibly that it outpaces the chips that train the best large language models (LLMs) like ChatGPT. In an ingenious turn, this hybrid technology also harnesses the relative energy efficiency of the human brain. An entire rack of these "computers" uses between 850-1,000W of energy versus 10,000-15,000W for a standard rack of AI chips.⁹ With plans to build datacenters exceeding five billion watts, a 10x more energy-efficient option could be a gamechanger.

Fungi and bacteria show promise as an organic building material

Scientists have recently developed a building material made of mycelium, which is the tubular, branching filament found in most fungi. They used a combination of red bread mold and *S. pasteurii*, a suitable bacterium, both of which are known for biomineralization, the same process that forms human bones and coral reefs.¹⁰

Why it matters:

Laying concrete currently accounts for 8% of humanity's carbon emissions each year. Finding a replacement material has the potential to eliminate concrete usage. Currently, the biomaterials created out of fungi and bacteria are not strong enough to support the same weight as concrete. But researchers are actively modifying the organisms to solve this problem, as well as incorporating traits like self-cleaning and self-repair.

Endnotes

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